

Creation and Implementation of Local and Cloud Virtual Machines

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Aim

To create and configure a local virtual machine using VMware and deploy a cloud-based virtual machine using Microsoft Azure.

Tools Required

- VMware Workstation
 - Ubuntu ISO
 - Microsoft Azure Portal
 - Internet Connection
 - SSH Client / Terminal
-

Cloud Virtual Machine (Azure)

1. Logged into the Microsoft Azure portal.
2. Created a new virtual machine by selecting the required OS and size.
3. Configured networking settings and security rules.
4. Launched the VM and obtained the public IP address.
5. Connected to the virtual machine remotely using SSH and verified its operation.

rpc-server - Microsoft Azure

https://portal.azure.com/?Microsoft_Azure_Education_correlationId=3d0a30a1-c57e-4ced-a247-ec031559a789&Microsoft_Azure_Educat...

Microsoft Azure Search resources, services, and docs (G+)

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rpc-server | Network settings

Virtual machine

Search

Diagnose and solve problems

Resource visualizer

Connect

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Availability + scale

Security

Network security group **rpc-server-nsg** (attached to networkInterface: **rpc-server964_z1**)
Impacts 0 subnets, 1 network interfaces

+ Create port rule

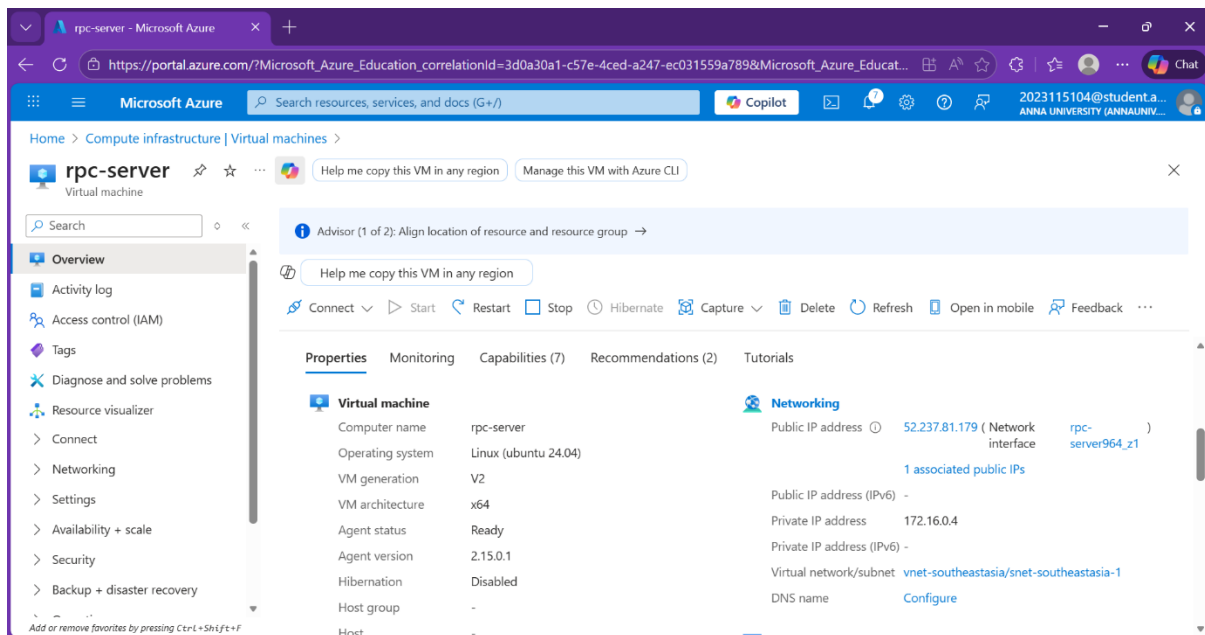
Search rules

Source == all Destination == all Protocol == all Action == all Port == all

Prio...	Name	Port	Protocol	Source	Destination
Inbound port rules (7)					
300	SSH	22	TCP	Any	Any
320	HTTP	80	TCP	Any	Any
340	HTTPS	443	TCP	Any	Any
1000	allow_rpc	8000	TCP	Any	Any
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork
65001	AllowAzureLoadBalancerInB...	Any	Any	AzureLoadBalancer	Any
65500	DenyAllInBound	Any	Any	Any	Any

```
azureuser@rpc-server: ~  
System information as of Thu Jan 29 17:39:32 UTC 2026  
  
System load: 0.08      Processes: 135  
Usage of /: 5.6% of 28.02GB  Users logged in: 0  
Memory usage: 3%      IPv4 address for eth0: 172.16.0.4  
Swap usage: 0%  
  
Expanded Security Maintenance for Applications is not enabled.  
0 updates can be applied immediately.  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
azureuser@rpc-server:~$ nano server.py  
azureuser@rpc-server:~$ python3 server.py  
RPC Server running on Azure VM at port 8000
```

```
azureuser@rpc-server: ~  
kavip@KaviPrakash:~$ ssh -i ~/rpc-server_key.pem azureuser@52.237.81.179  
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1017-azure x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Thu Jan 29 17:39:32 UTC 2026  
  
System load:  0.08      Processes:      135  
Usage of /:   5.6% of 28.02GB  Users logged in: 0  
Memory usage: 3%      IPv4 address for eth0: 172.16.0.4  
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```

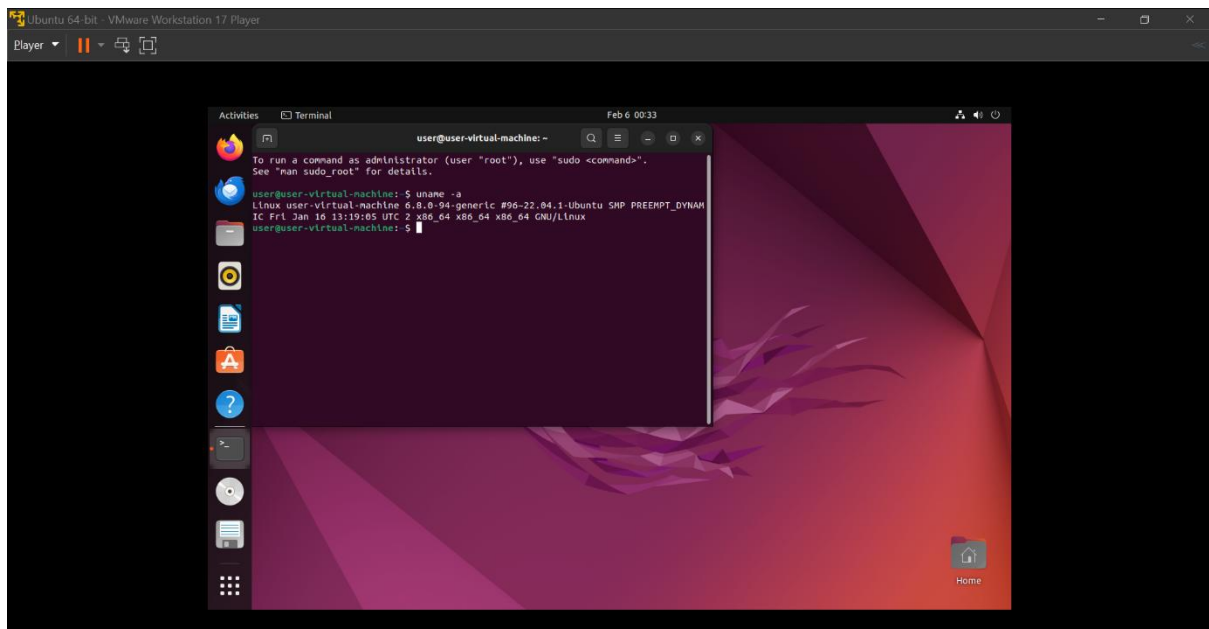


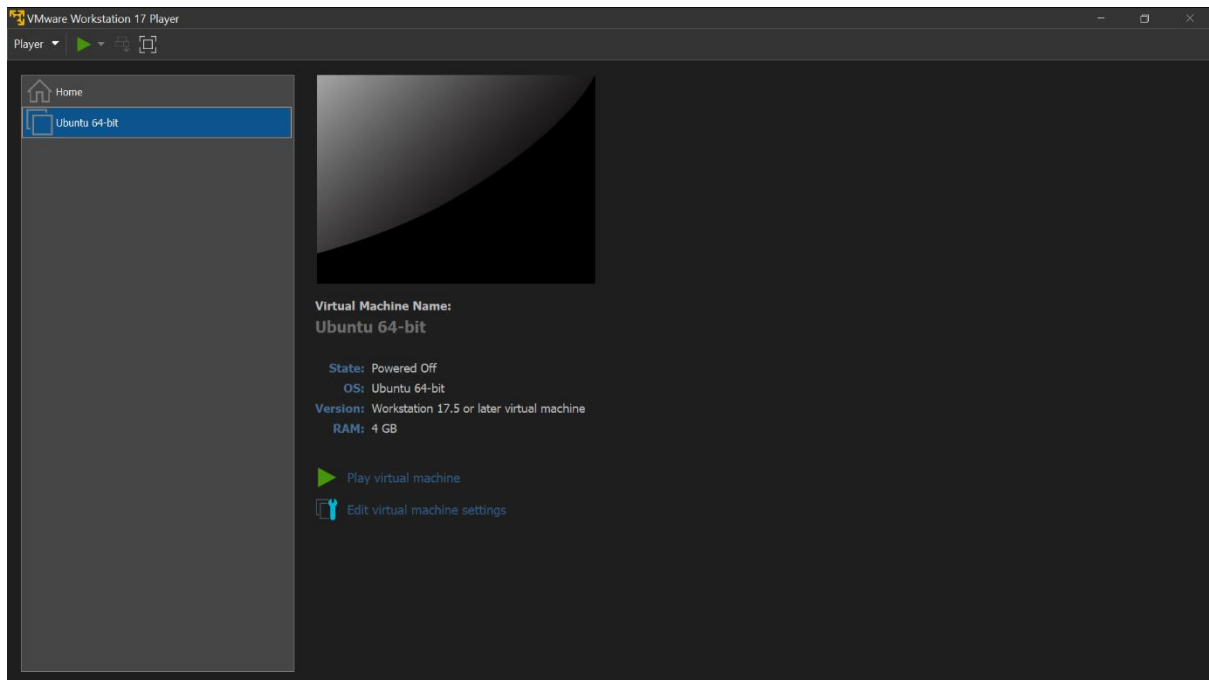
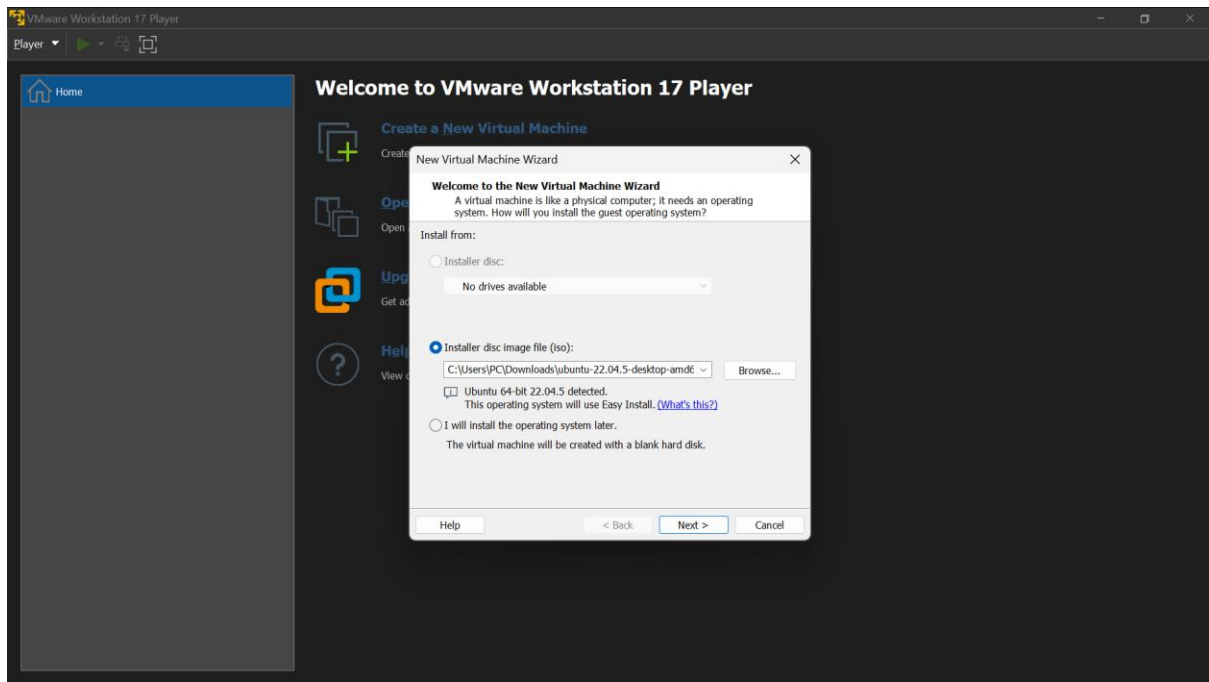
Procedure

Local Virtual Machine (Ubuntu)

1. Installed VMware Workstation and launched the application.

2. Created a new virtual machine and selected the Ubuntu ISO file.
3. Allocated system resources such as RAM, processor, and storage.
4. Installed the Ubuntu operating system by following the setup instructions.
5. Logged into the system and verified functionality using terminal commands.





Result

Both local and cloud virtual machines were successfully created, configured, and accessed. The Ubuntu VM operated on VMware, while the Azure VM was deployed and managed through the cloud.

Conclusion

This experiment helped in understanding virtualization concepts by implementing both local and cloud-based virtual machines, demonstrating their importance in modern computing environments.
